

INFINITY FITNESS STUDIOS

Data Dictionary

Member Retention, Utilization & Operational Analytics

December 2025 - February 2026

Document Control	Details
Document ID	IFS-DD-001
Version	1.0
Purpose	Define the analytical data entities and minimum fields required to support the confirmed Infinity Fitness analysis.
Source Format	Excel
Historical Coverage	Approximately one year
Confirmed Domains	Members, membership plans, payments, attendance/check-ins, joining/renewal dates, trainers, class bookings, revenue and expenses.
Field Naming Control	Original source-column names are not retained. Field names in this document are logical portfolio names, not claims about the client's exact Excel headers or database schema.
Confidentiality	No real member identifiers, names, financial values or other restricted client data are included.

1. Data Model Overview

The engagement used Excel-based operational data that required cleaning before analysis. The dictionary below reconstructs the minimum logical data structure needed to explain the completed retention, cohort, utilization, renewal, trainer and commercial analyses. It intentionally distinguishes confirmed business data from portfolio-standardized field names.

Logical Entity	Confirmed Business Content	Primary Analytical Use
Member	Member information	Member counts, segmentation and linkage
Membership	Plan, joining and renewal information	Tenure, retention, renewals, plan analysis
Attendance	Member attendance / check-ins	Engagement, attrition evidence, peak/underutilized periods
Payment	Member payment information	Membership/commercial analysis
Trainer	Trainer information and member assignments	Trainer-utilization analysis
Class / Session	Class bookings and trainer sessions	Bookings, workload and attendance analysis
Revenue	Revenue information	Commercial performance
Expense	Expense information	Expense analysis

2. Member Entity

Logical Field	Type	Definition / Use	Status
Member_ID	Identifier	Logical unique member key used to connect member activity across analytical datasets.	Required logical field; exact source name not retained
Member_Name	Text	Member name where present in source data; not required in public portfolio outputs.	Source detail not disclosed

Member_Status	Text	Membership/member status if supplied; may support active/cancelled interpretation.	Availability/rules to verify
Joining_Date	Date	Date the member joined; used for joining-month cohorts and tenure analysis.	Confirmed data element
Renewal_Date	Date	Membership renewal date where available; used for renewal analysis.	Confirmed data element

3. Membership / Plan Entity

Logical Field	Type	Definition / Use	Status
Membership_ID	Identifier	Logical membership record key if the source distinguishes membership records from members.	Logical design; exact source field not retained
Member_ID	Identifier	Links the membership to the relevant member.	Required analytical linkage
Plan_Type	Text	Membership offering associated with the record.	Confirmed concept
Membership_Start_Date	Date	Start date for the relevant membership period where available.	Logical analytical field

Membership_End_Date	Date	End/expiry date where available.	Availability to verify
Renewal_Date	Date	Renewal date/evidence associated with the membership.	Confirmed data element
Cancellation_Status_or_Date	Text/Date	Cancellation evidence if captured in the source.	Exact source representation not retained

Confirmed plan categories: monthly, 3-month, 6-month, annual and personal training.

4. Attendance / Check-in Entity

Logical Field	Type	Definition / Use	Status
Attendance_ID	Identifier	Logical unique attendance/check-in record identifier.	Logical design
Member_ID	Identifier	Member associated with the check-in.	Required analytical linkage
Checkin_Date	Date	Date of attendance/check-in.	Confirmed attendance concept; exact header not retained
Checkin_Time	Time	Time of attendance/check-in used for hour/day utilization analysis.	Required for time-of-day analysis; exact header not retained
Attendance_Status	Text	Valid/attended status where the source distinguishes attendance outcomes.	Availability to verify
Derived_Day	Text	Day/day-of-week derived from the check-in date for utilization analysis.	Derived analytical field
Derived_Hour	Number/Text	Hour/time band derived from check-in time for peak-period analysis.	Derived analytical field

5. Payment Entity

Logical Field	Type	Definition / Use	Status
Payment_ID	Identifier	Logical payment transaction identifier.	Logical design

Member_ID	Identifier	Member associated with the payment where linkable.	Confirmed analytical relationship; exact field name not retained
Payment_Date	Date	Date associated with the payment.	Logical analytical field
Payment_Amount	Decimal	Recorded payment amount.	Confirmed payment data; exact field name not retained
Payment_Category	Text	Payment classification where available, such as membership-related payment.	Exact source categories not retained
Plan_Type	Text	Plan association where present or derivable.	Availability/linkage to verify

6. Trainer Entity

Logical Field	Type	Definition / Use	Status
Trainer_ID	Identifier	Logical trainer key used to connect trainer activity.	Required logical field
Trainer_Name	Text	Trainer name where present; may be excluded from public portfolio outputs.	Confirmed trainer information; exact source header not retained
Member_ID	Identifier	Member assignment/linkage where trainer-member assignments are recorded.	Confirmed analytical concept
Assignment_Start/Period	Date/Period	Assignment period where available.	Availability to verify

7. Class / Session Entity

Logical Field	Type	Definition / Use	Status
Session_or_Booking_ID	Identifier	Logical record identifier for a session or class booking.	Logical design
Trainer_ID	Identifier	Trainer associated with the session/booking.	Required for trainer analysis where linkable
Member_ID	Identifier	Member associated with booking/session where applicable.	Required where member-level linkage exists
Session_Date	Date	Date of session/class activity.	Logical analytical field
Booking_Status	Text	Booking status where available.	Availability/rules to verify
Attendance_Status	Text	Attendance outcome where captured.	Confirmed attendance analysis concept; exact coding not retained
Session_Count	Number	Derived count of valid session records.	Derived analytical field

8. Revenue & Expense Entities

Entity	Logical Field	Type	Definition / Use	Status
Revenue	Revenue_Date	Date	Date/period associated with recorded revenue.	Logical analytical field
Revenue	Revenue_Amount	Decimal	Validated recorded revenue amount.	Confirmed revenue data; values confidential
Revenue	Revenue_Category	Text	Revenue classification where available.	Exact categories not retained
Expense	Expense_Date	Date	Date/period associated with recorded expense.	Logical analytical field
Expense	Expense_Amount	Decimal	Validated recorded expense amount.	Confirmed expense data; values confidential
Expense	Expense_Category	Text	Expense classification where available.	Exact categories not retained

9. Derived Analytical Fields

Derived Field	Derivation Intent	Used For
Joining_Month	Month/period derived from Joining_Date.	Cohort grouping
Membership_Duration	Elapsed duration between relevant membership/joining date and analysis/end/activity point, subject to agreed logic.	Duration segmentation
Retention_Flag	Indicates whether a member meets the agreed retained-	Retention analysis

	member rule for a reporting period.	
Attrition_Flag	Indicates whether a member meets the agreed stopped-attendance and/or cancellation rule.	Attrition analysis
Renewal_Flag	Indicates whether an eligible membership was renewed under the agreed rule.	Renewal rate
Attendance_Frequency	Count of valid member check-ins over the selected period.	Usage / engagement
Time_Band	Hour or grouped time period derived from check-in time.	Peak and underutilized-period analysis
Cohort_Period_Index	Relative period after the member's joining cohort.	Cohort retention tracking

10. Key Logical Relationships

From	To	Relationship Intent
Member	Membership	A member may have one or more membership records over time, depending on source structure.
Member	Attendance	A member may have multiple attendance/check-in records.
Member	Payment	A member may have multiple payment records.
Member	Trainer	Member-to-trainer assignment may exist where recorded.
Trainer	Class / Session	A trainer may be associated with multiple sessions/bookings.

Member	Class / Session	A member may have multiple bookings/sessions where captured.
Membership	Plan Type	Each membership record is associated with a plan classification where available.

11. Data Quality Controls

Control Area	Required Treatment
Identifiers	Check member/trainer identifiers for blanks, duplicates or inconsistent representation before joins/counts.
Duplicates	Identify duplicate member, attendance, payment, booking or session records that could inflate metrics.
Dates	Standardize joining, renewal, attendance, payment, session, revenue and expense dates before period analysis.
Plan Labels	Standardize equivalent membership-plan labels before plan comparison.
Missing Values	Do not automatically interpret missing renewal/cancellation/activity evidence as churn.
Amounts	Validate numeric revenue/payment/expense fields before aggregation.
Time Fields	Validate check-in time before hour/time-band analysis.
Cross-Entity Links	Validate key relationships sufficiently to avoid unmatched or duplicated analytical records.

12. Data-to-KPI Traceability

KPI Area	Primary Logical Fields
Retention	Member_ID, Joining_Date, membership/renewal evidence, activity evidence
Attrition	Member_ID, attendance activity, cancellation evidence where available
Cohort Retention	Member_ID, Joining_Date, attendance/retention evidence, Joining_Month, Cohort_Period_Index

Renewals	Member_ID, Plan_Type, Renewal_Date/Renewal_Flag
Utilization	Member_ID, Checkin_Date, Checkin_Time, Derived_Day, Derived_Hour/Time_Band
Plan Performance	Member_ID/Membership_ID, Plan_Type, renewal and payment evidence
Trainer Utilization	Trainer_ID, Member_ID, session/booking records, attendance, assignments
Revenue / Expenses	Revenue_Date/Amount, Expense_Date/Amount and available categories

13. Known Unknowns / Governance

Exact original Excel column names are not retained.

Exact source-file count, row count and member count are not retained.

Exact SQL table names, joins and query logic are not yet documented.

Exact inactivity window for stopped-attendance churn is not retained.

Exact cancellation/status coding is not retained.

Exact revenue and expense classifications are not retained.

No synthetic client values should be presented as historical data in the public portfolio.

14. Review Record

Role	Name	Review Status	Date
Gym Manager	_____	_____	_____
Data Analyst	_____	Dictionary Prepared	_____